

Operator Discovery Report: transfer_bottleneck_escape_perturbation

- Operator type: perturbation.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic stagnation escape tuned for bottleneck/two-cluster transfer: uses structured double-bridge as primary, with escalating segment reversals to rewire long inter-cluster connections, then a fixed fallback mode
- Novelty classification: stagnation_escalator.
- Rediscovery signature: perturbation:double_bridge:segment_reversal:3:3.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Track stagnation count during local search and restarts.
- Use double_bridge by default and escalate toward segment_reversal after the stagnation threshold.
- Increase perturbation strength deterministically when escape pressure rises.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 3749d73702a3acc7.

Why It Should Help

- The operator targets local-minimum escape directly by controlling when and how restarts are perturbed.

Failure Cases

- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 57.797957, distance-eval delta 0.0.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 579.211086, distance-eval delta 0.0.
- random_restart_2opt gap delta 0.0, runtime delta 6.832771, distance-eval delta 462.285714.
- sparse_three_opt gap delta 0.0, runtime delta -3.702957, distance-eval delta 0.0.
- clustered_local_search gap delta 0.017532, runtime delta -90.132443, distance-eval delta -3505.714286.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.0.
- Runtime inflation: 0.003318.
- Distance-evaluation inflation: 0.0.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.0.
- simplified_perturbation mean gap 0.121317 and delta vs full operator 0.0.
- alternate_perturbation mean gap 0.121317 and delta vs full operator 0.0.